

Research Methodology

Meaning, objectives, motivation of research, concept of deductive and inductive theory, characteristics of scientific method, understanding of concept, construct, variable; research process

Unit 1: Introduction to Research Methodology (4 Hrs.)

Meaning of Research: Research in common parlance refers to a search for knowledge. One can also define research as a scientific and systematic search for **pertinent** information on a specific topic. In fact, research is an art of scientific investigation. The Advanced Learner's Dictionary of Current English lays down the meaning of research as "a careful investigation or inquiry especially through search for new facts in any branch of knowledge" Redman and Mory define research as a "systematized effort to gain new knowledge." Some people consider research as a movement, a movement from the known to the unknown. It is actually a **voyage** of discovery. We all possess the vital instinct of inquisitiveness for, when the unknown confronts us, we wonder and our inquisitiveness makes us probe and attain full and fuller understanding of the unknown. This inquisitiveness is the mother of all knowledge and the method, which man employs for obtaining the knowledge of whatever the unknown, can be termed as research.

Research is an academic activity and as such the term should be used in a technical sense. According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organizing and evaluating data; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis. D. Slesinger and M. Stephenson in the Encyclopedia of Social Sciences define research as "the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art." Research is, thus, an original contribution to the existing stock of knowledge making for its advancement. It is the **pursuit** of truth with the help of study, observation, comparison and experiment. In short, the search for knowledge through objective and systematic method of finding solution to a problem is research. The systematic approach concerning generalization and the formulation of a theory is also research. As such the term 'research' refers to the systematic method consisting of enunciating the problem, formulating a hypothesis, collecting the facts or data, analyzing the facts and reaching certain conclusions either in the form of solution(s) towards the concerned problem or in certain generalizations for some theoretical formulation.

Definition: Research is a scientific inquiry aimed at learning new facts, testing ideas, etc. It is the systematic collection, analysis and interpretation of data to generate new knowledge and answer a certain question or solve a problem.

Question 1: Explain the meaning of research and define research.

Objectives:

The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. Though each research study has its own specific purpose, we may think of research objectives as falling into a number of following broad groupings:

1. To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as exploratory or formulate research studies)
2. To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies)

3. To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies)
4. To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies).

The objective may be termed as Broad Objective and Specific Objective.

Broad Objective:

The broad objective of the survey is to find out the proportion of population having access to population education and the proportion of those who knew about different media of population education. It is known that all media of population education are not equally accessible to the population. Therefore, the other objective is to identify the media which has been most effective in dissemination of knowledge about population education.

Specific Objective:

The specific objectives formulated are

- (i) To determine the problem proportion of population who have the knowledge about population education in Nepal by Age, Sex, and Ecological regions.
- (ii) To determine the components of population education commonly understood by Age, Sex and their Educational attainments of the respondents.
- (iii) To identify the media that has been most effective in dissemination of population education in view of respondents by Age, Sex and their Educational Attainment.

Question 2: Define research objectives. Discuss its types. Discuss how the research objectives are related to the research questions?

Motivation of Research:

What makes people to undertake research? This is a question of fundamental importance. The possible motives for doing research may be either one or more of the following:

1. Desire to get a research degree along with its consequential benefits.
2. Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates' research.
3. Desire to get intellectual joy of doing some creative work.
4. Desire to get respectability.
5. Desire to be service society.
6. Receiving fellowship from the government and private agencies.
7. Getting in depth knowledge about our interest subject.

However, this is not an exhaustive list of factors motivating people to undertake research studies. Many more factors such as directives of government, employment conditions, curiosity about new things, desire to understand causal

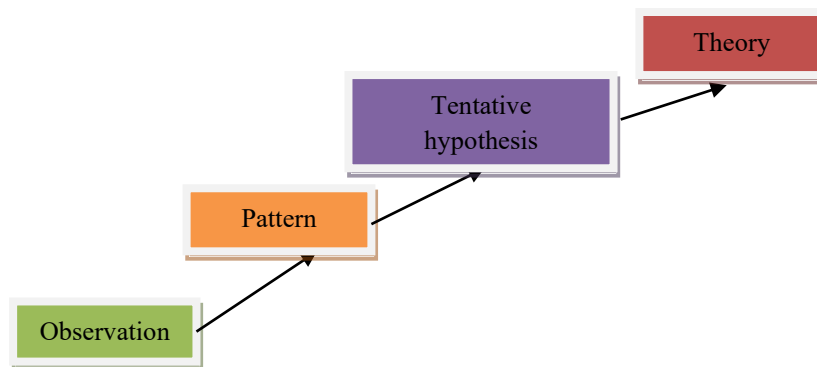
relationships, social thinking and awakening, and the like may as well motivate (or at times compel) people to perform research operations.

Question 3: Discuss the motivation of research.

Concept of deductive and inductive theory:

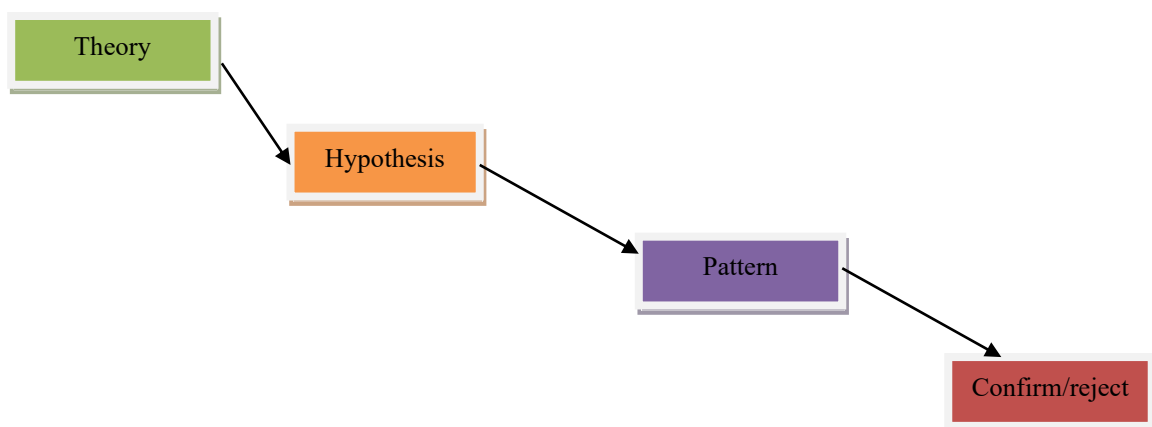
Inductive Theory:

Inductive theory is an approach to research that starts with the observation and the end result of the research is theory. Inductive approach to research does not have any place for hypothesis. So, researcher is free to alter direction of the study. Inductive approach is less structured as there is no guiding factor. Inductive approach is appropriate for small sample project because small numbers of sample are analyzed with greater gravity. More scrutinized information about the subject matter. Inductive approach assumes that there is abundance of time; so depth analysis is possible. Inductive approach is concerned with building a new theory. Inductive approach to research is bottom to up approach. The main principle of inductive method are specific to general, example to theory, concrete to abstract, method of discovery theory.



Deductive Theory:

Deductive theory is an approach to research that starts with the proposition of hypothesis and the end result of the research is confirmation/rejection. Deductive approach to research starts with the hypothesis itself. Researcher is bound not to alter direction of the study. Deductive approach is highly structured because there is some specific aim to be accomplished. Deductive approach is appropriate for large sample project because large numbers of sample are analyzed. Less scrutinized in getting information regarding subjects matter. Deductive approach assumes that there is shortage of time, so depth analysis of large. Deductive approach is concerned with testing an existing theory. Deductive approach to research is top to bottom approach. The main principle of deductive method are general to specific, theory to example, abstract to concrete, method of satisfied of theory.



Question 4: What do you mean by inductive and deductive theory?

Characteristics of scientific method:

The chief characteristics of scientific method are discussed in the following points.

- Objectivity
- Generality
- Verifiability
- Predictability
- Systematic
- Definiteness

Objectivity:

Objectivity is the property of objective reality, based on facts and existing outside the mind, not influenced by personal opinion or feeling. If there is not any place for personal beliefs, perception, biases, values, attitudes and emotions, then the single word to express this is the objectivity. Objectivity of the study concerns to (a) collect actual facts (b) Maximum impartially and (c) deal with naked facts.

To behave the properties of objectivity the scientific work must be always free of subjective ideas or feelings. It should be free from human bias and ought to be carried out or studied as it exists and perceived by nature and real surroundings.

Generality:

The characteristic of generalization of the result of a study to other subjects is generality. The laws derived from this method are universal (in the same of statistical universe) in application. The central interest of generality concern more in case of applied studies. The result derived from scientific method can be generalized for same universe. Generally refers to the facts that the scientific method must be a complete and universal. It should be applicable to almost all proved that every other things or circumstances remaining constant.

Verifiability

The conclusion drawn through a scientific method is subjected to verification at any time. The proposition is that the phenomenon under investigation must be capable of being observed and measured. In case direct observation could not be done, other methods such as interview can be utilized for verification. For instance, a man's order of preference of various jobs, although incapable of being observed can still be verified by means of an interview.

Predictability:

Predictability is the quality of predicting the result of a phenomenon. In scientific studies, the results can be predicted with sufficient accuracy. The cause and effect of observable fact related to two or more variable can be predicted in this method.

Studied done by scientific method should always be predictable about its basic nature and proceeding and finding because the similar phenomena or events under same condition and caused under the same method invariably must have the similar results, if not affected by other factors.

Systematic:

In every scientific study, there is an accepted mode of investigation. The result arrived by means of a haphazard method, even true, cannot be called scientific because its accuracy is purely accidental.

Definiteness:

It means the absence of unclearness and ambiguity. The study through scientific method gives rigid, definite conclusion. In this method, the confusing results are not concluded and clear, confined result is derived. In scientific method, if the experiment is performed again in order to check and increase the accuracy and precision of the result, the results must be definite and unambiguous. Thus the scientific method also can be said to follow the property of **repeatability**.

Question 5: Write the characteristics of scientific method and describe it.

Concept of research:

- In scientific research, concepts are abstract ideas or phenomena that are being studied (e.g. educational achievement).
- Conceptual research is defined as a methodology wherein research is conducted by observing and analyzing already present information on a given topic. Conceptual research doesn't involve conducting any practical experiments. It is related to abstract concepts or ideas. Philosophers have long used conceptual research to develop new theories or interpret existing theories in a different light.
- The most famous example of conceptual research is Sir Issac Newton. He observed his surroundings to conceptualize and develop theories about gravitation and motion.
- Einstein is widely known and appreciated for his work on conceptual research. Although his theories were based on conceptual observations, Einstein also proposed experiments to come up with theories to test conceptual research.

Advantage of conceptual research:

- Conceptual research mainly focuses on the concept of the research or the theory that explains a phenomenon. What causes the phenomenon, what are its building blocks, and so on? It's research based on pen and paper.
- This type of research heavily relies on previously conducted studies, no from of the experiment is conducted which saves time, effort, and resources. More relevant information can be generated by conducting conceptual research.
- Conceptual research is considered the most convenient form of research. In this type of research, if the conceptual framework is ready only relevant information and literature need to be sorted.

Constructs:

- Constructs can be conceptually defined in that they have meaning in theoretical terms. They can be abstract and do not necessarily need to be directly observable. Examples of constructs include intelligence or life satisfaction.
- Constructs are considered latent variables because they cannot be directly observable how and why certain phenomena behave the way that they do.

Variables:

- A variable in research simply refers to a person, place, thing, or phenomenon that you are trying to measure in some way.
- In research, variables are any characteristics that can take on different values, such as height, age, temperature, or test scores.
- A variable is, as the name applies, something that varies.
- Variable is the thing that keeps on changing. Examples; Weight, Height, Income, etc.
- Every concept can be variable, but every variable cannot be a concept.
- Variable is a concept that varies according to the individual and situation.

Question 6: What do you mean by concept of research? What is the advantage of concept of research?

Research process:

Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps. The following order concerning various steps provides a useful procedural guideline regarding the research process.

1. Formulating the research problem
2. Extensive literature survey
3. Developing the hypothesis
4. Preparing the research design
5. Determining sample design
6. Collecting the data
7. Execution of the project
8. Analysis of data
9. Hypothesis testing
10. Generalizations and interpretation, and
11. Preparation of the report or presentation of the results, i.e., formal write-up of conclusions reached.

Question 7: Write the steps involved in research process?

Questions:

1. Explain the meaning of research and define research.
2. Define and discuss the meaning of research.
3. What do you mean by research methodology?
4. Define research objectives. Discuss its types. Discuss how the research objectives are related to the research questions?
5. Discuss the motivation of research.
6. What do you mean by inductive and deductive theory?
7. Describe concept of inductive theory and deductive theory.
8. Write the characteristics of scientific method and describe it.
9. Describe the characteristics of scientific method.

10. What do you mean by concept of research? What is the advantage of concept of research?
11. Write the steps involved in research process?